



Top 35 Most Asked JavaScript Interview Q&A

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JavaScript Interview Questions & Answers

1- What is JavaScript?

Answer: JavaScript is a client-side scripting language used to make web pages interactive.

2- var, let, const difference?

var: Function-scoped, hoisted

let: Block-scoped

const: Block-scoped, cannot be reassigned

3- What is hoisting?

Answer: Variables and function declarations are moved to the top of their scope before execution.

4- What is the DOM?

Answer: Document Object Model — JavaScript's way of interacting with and manipulating HTML elements.

5- What are events?

Answer: Actions that happen in the browser (like click, mouseover). JavaScript can respond to these.

6- Difference between '==' and '==='?

'==' allows type coercion, while '===' requires both type and value to be the same.

7- Arrow functions?

Arrow functions are a shorthand way to write functions using the `=>` syntax.

```
const add = (a, b) => a + b;
```

8- What is the 'this' keyword in JavaScript?

'this' refers to the object that is executing the current function.

9- What is hoisting in JavaScript?

Hoisting is JavaScript's default behavior of moving declarations to the top.

10- What is a callback function?

Answer: A function passed into another function as an argument, to be executed later.

11- What are closures?

A closure is a function that has access to its own scope, the outer function's scope, and the global scope.

EX-

```
function outerFunction(outerVariable) {  
  function innerFunction(innerVariable) {  
    console.log(outerVariable);  
    console.log(innerVariable);  
  }  
  return innerFunction;  
}  
  
const myClosure = outerFunction('Hello');  
myClosure('World'); // Output: Hello, World
```

Explanation:

`outerFunction` takes `outerVariable` as an argument and returns `innerFunction`.

`innerFunction` has access to `outerVariable` even after `outerFunction` has finished executing.

When `myClosure` is called, it logs both `outerVariable` and `innerVariable`.

12- What is event bubbling in JavaScript?

Event bubbling is when an event triggers first on the innermost element and then propagates up to outer elements.

`event.stopPropagation();` // used to stop bubbling

13- Explain the difference between synchronous and asynchronous JavaScript.

Synchronous: Executes line by line (blocking).

Asynchronous: Uses callbacks/promises/async-await to execute non-blocking code (e.g., API calls).

14- What is a promise in JavaScript?

A Promise is an object representing the eventual completion (or failure) of an asynchronous operation.

```
let promise = new Promise((resolve, reject) => {  
  // async task  
});
```

15- What is the use of the async/await keywords?

They are used to write asynchronous code that looks synchronous.

16- What is the difference between null and undefined?

null is an assignment value, while undefined means a variable has been declared but not assigned.

17- What is the event loop in JavaScript?

The event loop handles the execution of multiple chunks of code over time.

18 - What is the difference between call, apply, and bind?

call and apply invoke functions immediately; bind returns a new function.

19- What is event delegation?

A technique where a single event listener handles events from multiple child elements using event bubbling

20- What is the spread operator?

Expands elements of an array or object.

```
let arr = [1, 2];
```

```
let newArr = [...arr, 3];
```

21- What is destructuring?

Destructuring is a way to extract values from arrays or properties from objects.

22- What is the difference between map() and forEach()?

map() returns a new array.

forEach() just iterates, no return.

23- What is localStorage vs sessionStorage?

localStorage: Data persists until manually deleted.

localStorage allows saving key/value pairs in a web browser with no expiration.

sessionStorage: Data cleared when session/tab closes.

24- What are higher-order functions?

Functions that take other functions as arguments or return functions.

25- What is JSON?

JSON is a format for storing and transporting data.

26- How does `setTimeout()` work?

The `setTimeout()` function in JavaScript schedules a function to be executed after a specified delay

`setTimeout(() => alert("Hi"), 2000);`

27- What are JavaScript data types?

Primitive: `string`, `number`, `boolean`, `null`, `undefined`, `symbol`, `bigint`

Non-primitive: `object`, `array`, `function`

28- What is a prototype?

A prototype is an object from which other objects inherit properties.

29- What is the difference between function declaration and expression?

A declaration is hoisted, an expression is not.

30- What is the use of the `typeof` operator?

`typeof` returns the type of a variable.

31- What are promises states?

`Pending`, `Fulfilled`, and `Rejected`.

32- What is a generator function?

A generator function can pause execution and resume later using `yield`.

33- What is the 'use strict' directive?

'use strict' helps catch common coding mistakes and prevents unsafe actions.

34- What is the difference between sync and async functions?

Sync functions block the thread, async functions do not

35- What is a Symbol in JavaScript?

Symbol is a primitive data type for unique identifiers.